RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

DEPARTMENT OF PHYSICAL SCIENCES

MAT 3222 – Numerical Analysis II

Tutorial 01

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1. Determine what type of solution exists, without graphing.

   a) $y = -\frac{3}{4}x + 2,\ 6x + 8y = 16$

   b) $4x - 2y = 7,\ y = 2x + 1$

   c) $3x + y = 8,\ 2x - y = 7$

2. Solve the system of equations by using the substitution method.

   a) $2x - y = 5,\ x + y = 2$

   b) $y = -x + 3,\ y = x + 1$

3. Solve the following questions using **Gaussian elimination without partial pivoting**:

   I. $3x + y - 2z = 6$
   $-3x + y + 2z = 8$
   $x - 2y + 4z = 12$

   II. $4x + y - 2z + w = 4$
   $-x - y + 3z + 2w = 5$

$-x + 3y - 2z - w = 12$

$3x + y + z + 4w = 10$

III. $x + y + z + w + v = 5$

$2x - y + z + w - v = 4$

$-x + 2y - z + 2w + v = 6$

$x + y + 2z - w + v = 3$

$2x - y + z + 3w + 2v = 10$

4. Solve the following questions using **Gaussian elimination with partial pivoting**

a) $x - y + z = 1$

$-3x + 2y - 3z = -6$

$2x - 5y + 4z = 5$

b) $2x + 4y - 2z - 2w = -4$

$x + 2y + 4z - 3w = 5$

$-3x - 3y + 8z - 2w = 7$

$-x + y + 6z - 3w = 7$

5. Solve the following questions using **Gauss - Jordan elimination**

i. $2x - y + 3z = 9$

$-x + 4y + z = 10$

$3x + 2y - z = 4$

ii. $2x + y - z + w = 8$

$x - y + z - 2w = 10$

$-x - 2y + z + w = 3$

$3x + y + z + 2w = 5$

Submit the tutorial on or before 1st of September (01/09/2026) at the tutor room.